PROBLEM SOLVING

Md. Rakibul Islam

ID: 223902078

Distance Between Two Points

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read the four values corresponding to the x and y axes of two points in the plane, p1 (x1, y1) and p2 (x2, y2) and calculate the distance between them, showing four decimal places, according to the formula:

Distance = $\sqrt{(x2 - x1)^2 + (y2 - y1)^2}$

Input

The input file contains two lines of data. The first one contains two double values: ***x1 y1*** and the second one also contains two double values with one digit after the decimal point: ***x2 y2***.

Output

Calculate and print the distance value using the provided formula, with 4 decimal places.

Input Sample	Output Sample
1.0 7.0 5.0 9.0	4.4721
-2.5 0.4 12.1 7.3	16.1484
2.5 -0.4 -12.2 7.0	16.4575

Distance

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Two cars (X and Y) leave in the same direction. The car X leaves with a constant speed of 60 km/h and the car Y leaves with a constant speed of 90 km / h.

In one hour (60 minutes) the car Y can get a distance of 30 kilometers from the X car, in other words, it can get away one kilometer for each 2 minutes.

Read the distance (in km) and calculate how long it takes (in minutes) for the car Y to take this distance in relation to the other car.

Input

The input file contains 1 integer value.

Output

Print the necessary time followed by the message "minutos" that means minutes in Portuguese.

Input Sample	Output Sample
30	60 minutos
110	220 minutos
7	14 minutos

Banknotes

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

In this problem you have to read an integer value and calculate the smallest possible number of banknotes in which the value may be decomposed. The possible banknotes are 100, 50, 20, 10, 5, 2 and 1. Print the read value and the list of banknotes.

Input

The input file contains an integer value **N** (0 < **N** < 1000000).

Output

Print the read number and the minimum quantity of each necessary banknotes in Portuguese language, as the given example. Do not forget to print the end of line after each line, otherwise you will receive *"Presentation Error"*.

Input Sample	Output Sample
576	576 5 nota(s) de R$ 100,00 1 nota(s) de R$ 50,00 1 nota(s) de R$ 20,00 0 nota(s) de R$ 10,00 1 nota(s) de R$ 5,00 0 nota(s) de R$ 2,00 1 nota(s) de R$ 1,00

Time Conversion

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read an integer value, which is the duration in seconds of a certain event in a factory, and inform it expressed in hours:minutes:seconds.

Input

The input file contains an integer **N**.

Output

Print the read time in the input file (seconds) converted in hours:minutes:seconds like the following example.

Input Sample	Output Sample
556	0:9:16
1	0:0:1
140153	38:55:53

Age in Days

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read an integer value corresponding to a person's age (in days) and print it in years, months and days, followed by its respective message “ano(s)”, “mes(es)”, “dia(s)”.

Note: only to facilitate the calculation, consider the whole year with 365 days and 30 days every month. In the cases of test there will never be a situation that allows 12 months and some days, like 360, 363 or 364. This is just an exercise for the purpose of testing simple mathematical reasoning.

Input

The input file contains 1 integer value.

Output

Print the output, like the following example.

Input Sample	Output Sample
400	1 ano(s) 1 mes(es) 5 dia(s)
800	2 ano(s) 2 mes(es) 10 dia(s)
30	0 ano(s) 1 mes(es) 0 dia(s)

beecrowd | 1021

Banknotes and Coins

By Neilor Tonin, URI Brazil

Timelimit: 1

Read a value of floating point with two decimal places. This represents a monetary value. After this, calculate the smallest possible number of *notes* and *coins* on which the value can be decomposed. The considered notes are of 100, 50, 20, 10, 5, 2. The possible coins are of 1, 0.50, 0.25, 0.10, 0.05 and 0.01. Print the message "NOTAS:" followed by the list of notes and the message "MOEDAS:" followed by the list of coins.

Input

The input file contains a value of floating point **N** (0 ≤ **N** ≤ 1000000.00).

Output

Print the minimum quantity of banknotes and coins necessary to change the initial value, as the given example.

Input Sample	Output Sample
576.73	NOTAS: 5 nota(s) de R$ 100.00 1 nota(s) de R$ 50.00 1 nota(s) de R$ 20.00 0 nota(s) de R$ 10.00 1 nota(s) de R$ 5.00 0 nota(s) de R$ 2.00 MOEDAS: 1 moeda(s) de R$ 1.00 1 moeda(s) de R$ 0.50

	0 nota(s) de R$ 10.00 1 nota(s) de R$ 5.00 0 nota(s) de R$ 2.00 MOEDAS: 1 moeda(s) de R$ 1.00 1 moeda(s) de R$ 0.50 0 moeda(s) de R$ 0.25 2 moeda(s) de R$ 0.10 0 moeda(s) de R$ 0.05 3 moeda(s) de R$ 0.01
4.00	NOTAS: 0 nota(s) de R$ 100.00 0 nota(s) de R$ 50.00 0 nota(s) de R$ 20.00 0 nota(s) de R$ 10.00 0 nota(s) de R$ 5.00 2 nota(s) de R$ 2.00 MOEDAS: 0 moeda(s) de R$ 1.00 0 moeda(s) de R$ 0.50 0 moeda(s) de R$ 0.25 0 moeda(s) de R$ 0.10 0 moeda(s) de R$ 0.05 0 moeda(s) de R$ 0.01
91.01	NOTAS: 0 nota(s) de R$ 100.00 1 nota(s) de R$ 50.00 2 nota(s) de R$ 20.00 0 nota(s) de R$ 10.00 0 nota(s) de R$ 5.00 0 nota(s) de R$ 2.00 MOEDAS: 1 moeda(s) de R$ 1.00 0 moeda(s) de R$ 0.50 0 moeda(s) de R$ 0.25 0 moeda(s) de R$ 0.10 0 moeda(s) de R$ 0.05 1 moeda(s) de R$ 0.01

beecrowd | 1035

Selection Test 1

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read 4 integer values A, B, C and D. Then if B is greater than C and D is greater than A and if the sum of C and D is greater than the sum of A and B and if C and D were positives values and if A is even, write the message **"Valores aceitos"** (Accepted values). Otherwise, write the message **"Valores nao aceitos"** (Values not accepted).

Input

Four integer numbers A, B, C and D.

Output

Show the corresponding message after the validation of the values.

Input Sample	Output Sample
5 6 7 8	Valores nao aceitos
2 3 2 6	Valores aceitos

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Bhaskara's Formula

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read 3 floating-point numbers. After, print the roots of bhaskara's formula. If it's impossible to calculate the roots because a division by zero or a square root of a negative number, presents the message *"Impossivel calcular"*.

Input

Read 3 floating-point numbers (double) A, B and C.

Output

Print the result with 5 digits after the decimal point or the message if it is impossible to calculate.

Input Samples	Output Samples
10.0 20.1 5.1	R1 = -0.29788 R2 = -1.71212
0.0 20.0 5.0	Impossivel calcular
10.3 203.0 5.0	R1 = -0.02466 R2 = -19.68408
10.0 3.0 5.0	Impossivel calcular

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Interval

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

You must make a program that read a float-point number and print a message saying in which of following intervals the number belongs: [0,25] (25,50], (50,75], (75,100]. If the read number is less than zero or greather than 100, the program must print the message “Fora de intervalo” that means "Out of Interval".

The symbol '(' represents greather than. For example:
[0,25] indicates numbers between 0 and 25.0000, including both.
(25,50] indicates numbers greather than 25 (25.00001) up to 50.0000000.

Input

The input file contains a floating-point number.

Output

The output must be a message like following example.

Input Sample	Output Sample
25.01	Intervalo (25,50]
25.00	Intervalo [0,25]
100.00	Intervalo (75,100]
-25.02	Fora de intervalo

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Snack

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Using the following table, write a program that reads a code and the amount of an item. After, print the value to pay. This is a very simple program with the only intention of practice of selection commands.

CODE	SPECIFICATION	PRICE
1	Cachorro Quente	R$ 4.00
2	X-Salada	R$ 4.50
3	X-Bacon	R$ 5.00
4	Torrada simples	R$ 2.00
5	Refrigerante	R$ 1.50

Input

The input file contains two integer numbers **X** and **Y**. **X** is the product code and **Y** is the quantity of this item according to the above table.

Output

The output must be a message "Total: R$ " followed by the total value to be paid, with 2 digits after the decimal point.

Input Sample	Output Sample
3 2	Total: R$ 10.00
4 3	Total: R$ 6.00
2 3	Total: R$ 13.50

Average 3

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read four numbers (N_1, N_2, N_3, N_4), which one with 1 digit after the decimal point, corresponding to 4 scores obtained by a student. Calculate the average with weights 2, 3, 4 e 1 respectively, for these 4 scores and print the message *"Media: "* (Average), followed by the calculated result. If the average was 7.0 or more, print the message *"Aluno aprovado."* (Approved Student). If the average was less than 5.0, print the message: *"Aluno reprovado."* (Reproved Student). If the average was between 5.0 and 6.9, including these, the program must print the message "Aluno em exame." (In exam student).

In case of exam, read one more score. Print the message *"Nota do exame: "* (Exam score) followed by the typed score. Recalculate the average (sum the exam score with the previous calculated average and divide by 2) and print the message *"Aluno aprovado."* (Approved student) in case of average 5.0 or more) or *"Aluno reprovado."* (Reproved student) in case of average 4.9 or less. For these 2 cases (approved or reproved after the exam) print the message *"Media final: "* (Final average) followed by the final average for this student in the last line.

Input

The input contains four floating point numbers that represent the students' grades.

Output

Print all the answers with one digit after the decimal point.

Input Sample	Output Sample
2.0 4.0 7.5 8.0 6.4	Media: 5.4 Aluno em exame. Nota do exame: 6.4 Aluno aprovado. Media final: 5.9
2.0 6.5 4.0 9.0	Media: 4.8 Aluno reprovado.
9.0 4.0 8.5 9.0	Media: 7.3 Aluno aprovado.

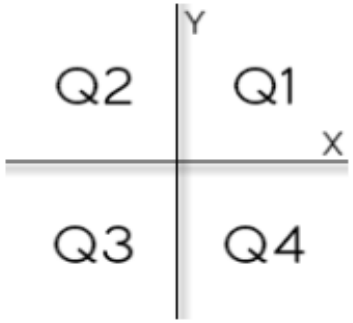
beecrowd | 1041

Coordinates of a Point

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Write an algorithm that reads two floating values (x and y), which should represent the coordinates of a point in a plane. Next, determine which quadrant the point belongs, or if you are at one of the Cartesian axes or the origin (x = y = 0).



If the point is at the origin, write the message "Origem".

If the point is at X axis write "Eixo X", else if the point is at Y axis write "Eixo Y".

Input

The input contains the coordinates of a point.

Output

The output should display the quadrant in which the point is.

Input Sample	Output Sample
4.5 -2.2	Q4
0.1 0.1	Q1

beecrowd | 1042

Simple Sort

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read three integers and sort them in ascending order. After, print these values in ascending order, a blank line and then the values in the sequence as they were readed.

Input

The input contains three integer numbers.

Output

Present the output as requested above.

Input Sample	Output Sample
7 21 -14	-14 7 21 7 21 -14
-14 21 7	-14 7 21 -14 21 7

beecrowd | 1044

Multiples

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read two integer values (A and B). After, the program should print the message **"Sao Multiplos"** (are multiples) or **"Nao sao Multiplos"** (aren't multiples), corresponding to the read values.

Input

The input has two integer numbers.

Output

Print the relative message to the input as stated above.

Input Sample	Output Sample
6 24	Sao Multiplos
6 25	Nao sao Multiplos

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Triangle Types

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read 3 double numbers (A, B and C) representing the sides of a triangle and arrange them in decreasing order, so that the side A is the biggest of the three sides. Next, determine the type of triangle that they can make, based on the following cases always writing an appropriate message:

- if $A \geq B + C$, write the message: **NAO FORMA TRIANGULO**
- if $A^2 = B^2 + C^2$, write the message: **TRIANGULO RETANGULO**
- if $A^2 > B^2 + C^2$, write the message: **TRIANGULO OBTUSANGULO**
- if $A^2 < B^2 + C^2$, write the message: **TRIANGULO ACUTANGULO**
- if the three sides are the same size, write the message: **TRIANGULO EQUILATERO**
- if only two sides are the same and the third one is different, write the message: **TRIANGULO ISOSCELES**

Input

The input contains three double numbers, A ($0 < A$) , B ($0 < B$) and C ($0 < C$).

Output

Print all the classifications of the triangle presented in the input.

Input Samples	Output Samples
7.0 5.0 7.0	TRIANGULO ACUTANGULO TRIANGULO ISOSCELES
6.0 6.0 10.0	TRIANGULO OBTUSANGULO TRIANGULO ISOSCELES
6.0 6.0 6.0	TRIANGULO ACUTANGULO TRIANGULO EQUILATERO
5.0 7.0 2.0	NAO FORMA TRIANGULO
6.0 8.0 10.0	TRIANGULO RETANGULO

beecrowd | 1046

Game Time

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read the start time and end time of a game, in hours. Then calculate the duration of the game, knowing that the game can begin in a day and finish in another day, with a maximum duration of 24 hours. The message must be printed in portuguese "O JOGO DUROU X HORA(S)" that means "THE GAME LASTED X HOUR(S)"

Input

Two integer numbers representing the start and end time of a game.

Output

Print the duration of the game as in the sample output.

Input Sample	Output Sample
16 2	O JOGO DUROU 10 HORA(S)
0 0	O JOGO DUROU 24 HORA(S)
2 16	O JOGO DUROU 14 HORA(S)

beecrowd | 1047

Game Time with Minutes

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read the start time and end time of a game, in hours and minutes (initial hour, initial minute, final hour, final minute). Then print the duration of the game, knowing that the game can begin in a day and finish in another day,

Obs.: With a maximum game time of 24 hours and the minimum game time of 1 minute.

Input

Four integer numbers representing the start and end time of the game.

Output

Print the duration of the game in hours and minutes, in this format: "O JOGO DUROU XXX HORA(S) E YYY MINUTO(S)" . Which means: the game lasted XXX hour(s) and YYY minutes.

Input Sample	Output Sample
7 8 9 10	O JOGO DUROU 2 HORA(S) E 2 MINUTO(S)
7 7 7 7	O JOGO DUROU 24 HORA(S) E 0 MINUTO(S)
7 10 8 9	O JOGO DUROU 0 HORA(S) E 59 MINUTO(S)

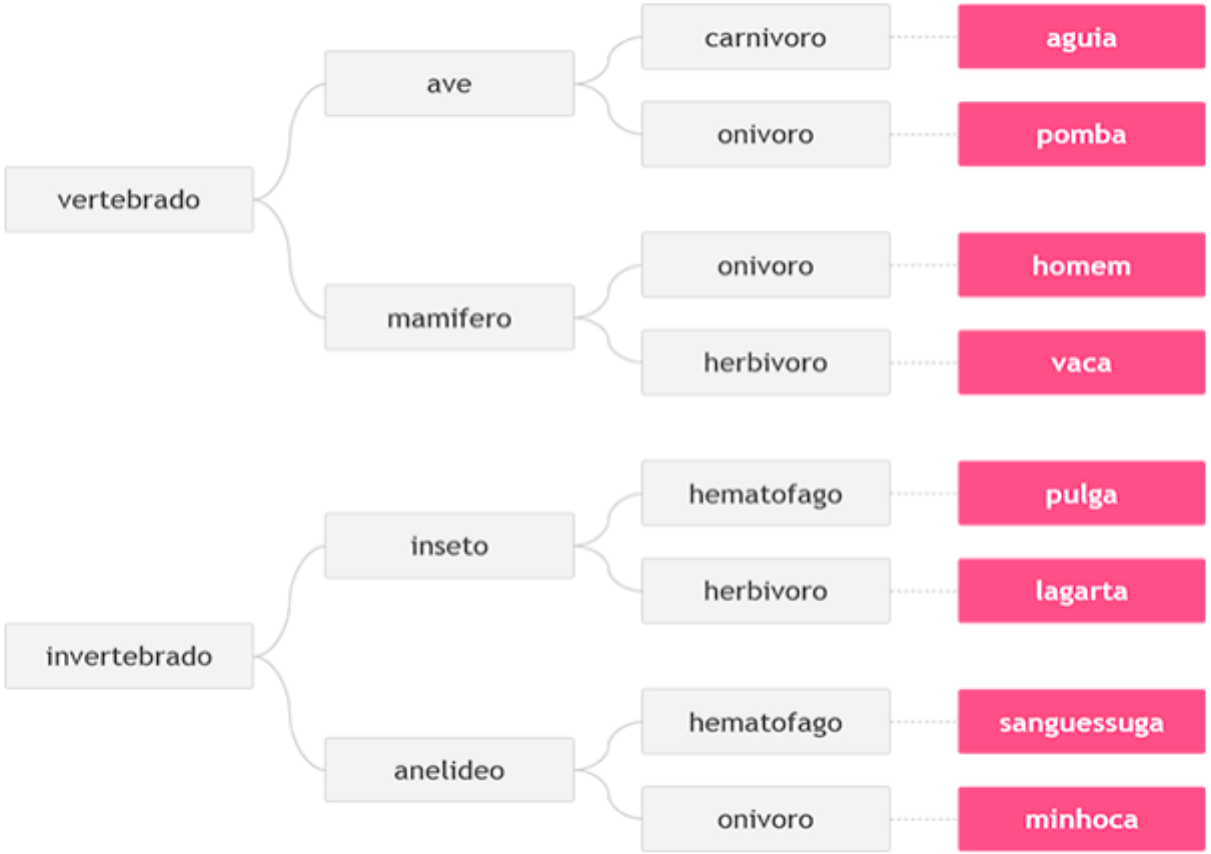
beecrowd | 1049

Animal

By Neilor Tonin, URI Brazil

Timelimit: 1

In this problem, your job is to read three Portuguese words. These words define an animal according to the table below, from left to right. After, print the chosen animal defined by these three words.



Input

The input contains 3 words, one by line, that will be used to identify the animal, according to the above table, with all letters in lowercase.

Output

Print the animal name according to the given input.

Input Samples	Output Samples
vertebrado mamifero onivoro	homem
vertebrado ave carnivoro	aguia
invertebrado anelideo onivoro	minhoca

beecrowd | 1050

DDD

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read an integer number that is the code number for phone dialing. Then, print the destination according to the following table:

DDD	Destination
61	Brasilia
71	Salvador
11	Sao Paulo
21	Rio de Janeiro
32	Juiz de Fora
19	Campinas
27	Vitoria
31	Belo Horizonte

If the input number isn't found in the above table, the output must be:
DDD nao cadastrado
That means "DDD not found" in Portuguese language.

Input

The input consists in a unique integer number.

Output

Print the city name corresponding to the input DDD. Print *DDD nao cadastrado* if doesn't exist corresponding DDD to the typed number.

Input Sample	Output Sample
11	Sao Paulo

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Taxes

By Neilor Tonin, URI Brasil

Timelimit: 1

In an imaginary country called Lisarb, all the people are very happy to pay their taxes because they know that doesn't exist corrupt politicians and the taxes are used to benefit the population, without any misappropriation. The currency of this country is Rombus, whose symbol is R$.

Read a value with 2 digits after the decimal point, equivalent to the salary of a Lisarb inhabitant. Then print the due value that this person must pay of taxes, according to the table below.

Salary	Taxes
from 0.00 to R$ 2,000.00	Without taxes
from R$ 2,000.01 to R$ 3,000.00	8 %
from R$ 3,000.01 to R$ 4,500.00	18 %
more than R$ 4,500.00	28 %

Remember, if the salary is R$ 3,002.00 for example, the rate of 8% is only over R$ 1,000.00, because the salary from R$ 0.00 to R$ 2,000.00 is tax free. In the follow example, the total rate is 8% over R$ 1000.00 + 18% over R$ 2.00, resulting in R$ 80.36 at all. The answer must be printed with 2 digits after the decimal point.

Input

The input contains only a float-point number, with 2 digits after the decimal point.

Output

Print the message "R$" followed by a blank space and the total tax to be payed, with two digits after the decimal point. If the value is up to 2000, print the message "Isento".

Input Samples	Outputs Samples
3002.00	R$ 80.36
1701.12	Isento
4520.00	R$ 355.60

beecrowd | 1052

Month

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read an integer number between 1 and 12, including. Corresponding to this number, you must print the month of the year, in english, with the first letter in uppercase.

Input

The input contains only an integer number.

Output

Print the name of the month according to the input number, with the first letter in uppercase.

Input Sample	Output Sample
4	April

Event Time

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Peter is organizing an event in his University. The event will be in April month, beginning and finishing within April month. The problem is: Peter wants to calculate the event duration in seconds, knowing obviously the begin and the end time of the event.

You know that the event can take from few seconds to some days, so, you must help Peter to compute the total time corresponding to duration of the event.

Input

The first line of the input contains information about the beginning day of the event in the format: "Dia xx". The next line contains the start time of the event in the format presented in the sample input. Follow 2 input lines with same format, corresponding to the end of the event.

Output

Your program must print the following output, one information by line, considering that if any information is null for example, the number 0 must be printed in place of W, X, Y or Z:

```
W dia(s)
X hora(s)
Y minuto(s)
Z segundo(s)
```

Obs: Consider that the event of the test case have the minimum duration of one minute. "dia" means day, "hora" means hour, "minuto" means minute and "Segundo" means second in Portuguese.

Input Sample	Output Sample
Dia 5 08 : 12 : 23 Dia 9 06 : 13 : 23	3 dia(s) 22 hora(s) 1 minuto(s) 0 segundo(s)

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Positives and Average

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read 6 values that can be floating point numbers. After, print how many of them were positive. In the next line, print the average of all positive values typed, with one digit after the decimal point.

Input

The input consist in 6 numbers that can be integer or floating point values. At least one number will be positive.

Output

The first output value is the amount of positive numbers. The next line should show the average of the positive values typed.

Input Sample	Output Sample
7 -5 6 -3.4 4.6 12	4 valores positivos 7.4

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Even Between five Numbers

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Make a program that reads five integer values. Count how many of these values are even and print this information like the following example.

Input

The input will be 5 integer values.

Output

Print a message like the following example with all letters in lowercase, indicating how many even numbers were typed.

Input Sample	Output Sample
7 -5 6 -4 12	3 valores pares

beecrowd | 1066

Even, Odd, Positive and Negative

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Make a program that reads five integer values. Count how many of these values are even, odd, positive and negative. Print these information like following example.

Input

The input will be 5 integer values.

Output

Print a message like the following example with all letters in lowercase, indicating how many of these values are even, odd, positive and negative.

Input Sample	Output Sample
-5 0 -3 -4 12	3 valor(es) par(es) 2 valor(es) impar(es) 1 valor(es) positivo(s) 3 valor(es) negativo(s)

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Odd Numbers

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read an integer value **X** (1 <= **X** <= 1000). Then print the odd numbers from 1 to **X**, each one in a line, including **X** if is the case.

Input

The input will be an integer value.

Output

Print all odd values between 1 and **X**, including **X** if is the case.

Input Sample	Output Sample
8	1 3 5 7

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Six Odd Numbers

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read an integer value **X** and print the 6 consecutive odd numbers from **X**, a value per line, including **X** if it is the case.

Input

The input will be a positive integer value.

Output

The output will be a sequence of six odd numbers.

Input Sample	Output Sample
8	9 11 13 15 17 19

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Sum of Consecutive Odd Numbers I

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read two integer values **X** and **Y**. Print the sum of all odd values between them.

Input

The input file contain two integer values.

Output

The program must print an integer number. This number is the sum off all odd values between both input values that must fit in an integer number.

Sample Input	Sample Output
6 -5	5
15 12	13
12 12	0

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Interval 2

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read an integer **N**. This N will be the number of integer numbers **X** that will be read.

Print how many these numbers **X** are in the interval [10,20] and how many values are out of this interval.

Input

The first line of input is an integer **N** (**N** < 10000), that indicates the total number of test cases.
Each case is an integer number **X** (-10^7 < **X** < 10^7).

Output

For each test case, print how many numbers are in and how many values are out of the interval.

Input Sample	Output Sample
4 14 123 10 -25	2 in 2 out

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Even Square

Adapted by Neilor Tonin, URI Brazil

Timelimit: 1

Read an integer **N**. Print the square of each one of the even values from 1 to **N** including **N** if it is the case.

Input

The input contains an integer **N** (5 < **N** < 2000).

Output

Print the square of each one of the even values from 1 to **N**, as the given example.

Be carefull! Some language automaticly print 1e+006 instead 1000000. Please configure your program to print the correct format setting the output precision.

Input Sample	Output Sample
6	2^2 = 4 4^2 = 16 6^2 = 36